Candidate Seat No:________________

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Science in Information Technology (M.Sc. (IT))

Semester-I University Examination November-2019

MS 112 Operating System Concepts

Date: 19.11.19, Tuesday Time: 10.00 a.m. to 01.00 p.m. Maximum Marks: 70

Instructions:

1. **Attempt both sections in separate answer books.**
2. **The figure to the right indicates marks.**
3. **Make a suitable assumption when necessary.**
4. **Simple Calculator is permitted.**

SECTION-1

Q.1 **Do as directed. (One mark for each question).** **[11]**

1. Define: critical section.
2. Give names of any two non-preemptive scheduling policy.
3. List out the states of five state model that resides in main memory.
4. What do you mean by virtual memory?
5. Which kind of ordering of resource types is needed to prevent circular wait condition of deadlock?
6. Which transition of seven state process model shows the poor design of the model?
7. Fixed partitioning leads to ____________________fragmentation.
8. List out the transitions of two state process model.
9. What is the decision mode of Round Robin policy?
10. ___________ provides an interface to the services made available by an operating system.
11. Define: Distributed Operating Systems.

Q.2 [A] What do you mean by semaphore? What are the characteristics of it?.Write a solution of producer consumer using semaphore. **[05]**

[B] Discuss the role of an operating system in the context of (a) program execution (b) I/O access (c) file access (d) program development and (e) system access. **[05]**

[C] Explain physical and logical organization requirements of memory management. **[02]**

OR

Q.2 [A] Discuss the both solutions of Dining philosopher problem using semaphore. **[05]**

[B] Explain paging concept of memory management with example. Differentiate paging with fixed partitioning and dynamic partitioning. **[05]**

[C] Explain the conditions of the deadlock. **[02]**

Q.3 **Answer the following questions in details**(**Any Three**). **[12]**

1. Explain Network Operating System in detail.
2. What do you mean by process? Explain Process Control Block in details.

[**P.T.O**]

3. Explain feedback scheduling policy with appropriate example.
4. Explain five state process model with all states and list out its limitation.
5. Determine the number of page faults according to Optimal and LRU replacement policies of virtual memory.
The sequence of pages for the execution of process :
1,3,2,1,5,4,1,2,5,3,2,1,3,2,4,1,2,4,3,1,3,2,4,5
The resident set of the process is 4.

SECTION-2

Q.4 **Do as directed. (One mark for each question).** **[11]**

1. Give the linux command that display users who use the linux.
2. Sensors are __________redable I/O device.
3. Define : Controller.
4. Give the disadvantages of the single level directory.
5. __________fragmentation occurs in contiguous allocation method.
6. State the principle of least privilege.
7. __________is the connection point through which the device communicates with machine.
8. ______is the Linux command to rename the file.
9. How to compare two files in Linux?
10.Variable allocation type of indexed allocation is similar to____________.
11. Give the difference between delete and truncate file operations.

Q.5 [A] Explain programmed I/O and Interrupt driven I/O techniques. **[04]**
[B] Write a shell script to find prime numbers between 1 to N. **[04]**
[C] Explain different layers of file system. **[04]**

OR

Q.5[A] Explain chained file allocation method and compare it with contiguous and indexed allocation. **[04]**
[B] Explain any four kernel I/O subsystems. **[04]**
[C] Give the characteristics of RAID architecture and explain any one layer having a true RAID characteristics. **[04]**

Q.6 **Answer following questions in detail** (**Any Three**). **[12]**

1. Explain security violations and list out different levels of security measure.
2. Explain **ls** command with at least eight switches.
3. Write a shell script to accept any N numbers from user and compute the multiplication of only odd numbers.Use appropriate validations for the same.
4. Explain the steps of file system mounting.
5.Explain any four commands of Linux related to process management.
